FACULTY OF ENGINEERING
CHULALONGKORN UNIVERSITY
2110201 Computer Engineering Mathematics
Year II, Second Semester, Final Examination, May 17, 2022 13:00-16:00

Name.....Student ID..... CR58 .....

Remarks

1. **There are 23 questions in this exam paper, with the total number of 12 pages including this page.**
2. **The exam duration is 180 minutes.**
3. Write your Student ID, full name in the space provided on the top of every page.
4. No documents or calculators are allowed inside the exam room except for non-programmable and non-graphic calculators.
5. Borrowing is not allowed unless it is supervised by the proctor.
6. You must not bring any part of this exam paper outside. The exam paper is a government's property. Violators will be prosecuted under a criminal court.
7. Student who wishes to leave the exam room before the end of the exam period, must raise his/her hand and ask for permission before leaving the room. Student must leave the room in the orderly manner.
8. Once the time is expired, student must stop writing and must remain seated quietly until the proctors collect all the exam papers or given exam booklets. Only then, the students will be allowed to leave the room in the orderly manner.
9. Any student who does not obey the regulations listed above will receive punishment under the Faculty of Engineering Official Announcement on January 6, 2003 regarding the exam regulations.
   a) **With implicit evidence or showing intention for cheating, student will receive an F in that subject and will receive an academic suspension for 1 semester.**
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**ห้ามนิสิตพกโทรศัพท์และอุปกรณ์สื่อสารไว้กับตัวระหว่างสอบ หากตรวจพบจะถือว่านิสิตกระทำผิดเกี่ยวกับการสอบ อาจต้องพ้นสภาพการเป็นนิสิต หรือ ให้ได้รับ F และอาจพิจารณาให้ถอนรายวิชาอื่นทั้งหมดที่ลงทะเบียนไว้ในภาคการศึกษานี้**

*** ร่วมรณรงค์การไม่กระทำผิดและไม่ทุจริตการสอบที่คณะวิศวกรรมศาสตร์ ***

**ข้าพเจ้ายอมรับในข้อกำหนดที่กล่าวมานี้ ข้าพเจ้าเป็นผู้ทำข้อสอบนี้ด้วยตนเองโดยมิได้รับการช่วยเหลือ หรือให้ความช่วยเหลือ ในการทำข้อสอบนี้**

ลงชื่อนิสิต.....

วันที่.....

1. Pencil is allowed.
2. Write your answer in the provided space only. If the space is not enough, you must use the back of the same page only.

Name_____ ID _____CR58_____

Part I: (25 points) Basic Skill, provide detailed answer

1. (4 points) Find the null space of the following matrix

$$\begin{bmatrix} 0 & 1 & 0 & 5 & 8 & 2 & 4 \\ 0 & 0 & 0 & 0 & 5 & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 & 3 & 1 \\ 0 & 0 & 0 & 0 & 7 & 2 & 3 \end{bmatrix}$$

2. (2 points) Let $S = \text{span}(\{v_1, v_2\})$ where $v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. Find $S^{\perp}$.

3. (4 points) Let $A$ be a $3 \times 4$ matrix such that

$$\text{RRE}(A) = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } \text{RRE}(A^T) = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Find a basis for the null space, column space, row space and left null space of A.
What is the rank and nullity of A?

4. (3 points) Find the orthonormal basis by applying Gram-Schmidt algorithm to the given vectors in the given order, i.e., the direction of $v_1$ should stay the same.
$v_1 = (1,1,0)$, $v_2 = (1,1,1)$, $v_3 = (1,0,0)$

5. (6 points) Find all eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$.

6. (6 points) An SVD of matrix $A$ is given by $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & 0.5 \end{bmatrix}^T$

Matrix $A$ has _____ rows and _____ columns.

Rank of $A$ = _____ Nullity of $A$ = ____

From the SVD, write down a basis of each of the following spaces

$row(A)$

$col(A)$

$nul(A)$

$nul(A^T)$

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Part II (2 points per question): Provide short answer only.

1. There is a square matrix $A$ such that $\det(A) = -1$. If $A$ has 6 columns and each column is represented by $v_i$ $(1 \leq i \leq 6)$, what is the minimum and maximum angle between any pair of $v_i$ and $v_j$?

   Maximum:

   Minimum:

2. A matrix $A$ has 2 rows and 3 columns. If $A^T(\vec{x} + t\vec{v}) = A^T(\vec{x})$ only when $\vec{v} = (0,0,0)$ what is the dimension of the nullspace of $A$?

   Answer:

3. Let $S$ be a subspace of $\mathbb{R}^4$, the dimension of the pre-image of $S$ with respect to a function $h$ is 3. If $h$ can be represented by a matrix $A$ and the column space of $A$ is $S$, write a basis of the null space of A that contains smallest number of vectors.

   Answer:

4. Let $W$ be a set of 10 linearly independent vectors. Each is a member of $\mathbb{R}^{15}$. If $A$ is a matrix that represent a distortion free mapping such that each column of $A$ must be a member of $\mathrm{perp}(W)$, what are the maximum and minimum size of $A$ (answer as a "$a \times b$" where $a$ and $b$ are positive integers.

   Maximum:

   Minimum:

5. Consider a non-standard orthonormal basis $B$ of $\mathbb{R}^3$ and let $W$ be a change-of-basis matrix such that $W\vec{x}$ is a representation of $\vec{x}$ with respect to $B$. There exist several sets of two linearly independent vectors $\{v_1, v_2\}$ such that $W\overrightarrow{v_1} = \overrightarrow{v_1}$ and $W\overrightarrow{v_2} = \overrightarrow{v_2}$. If the entries of $W$ is either -1, 0 or 1, write all possible $W$.

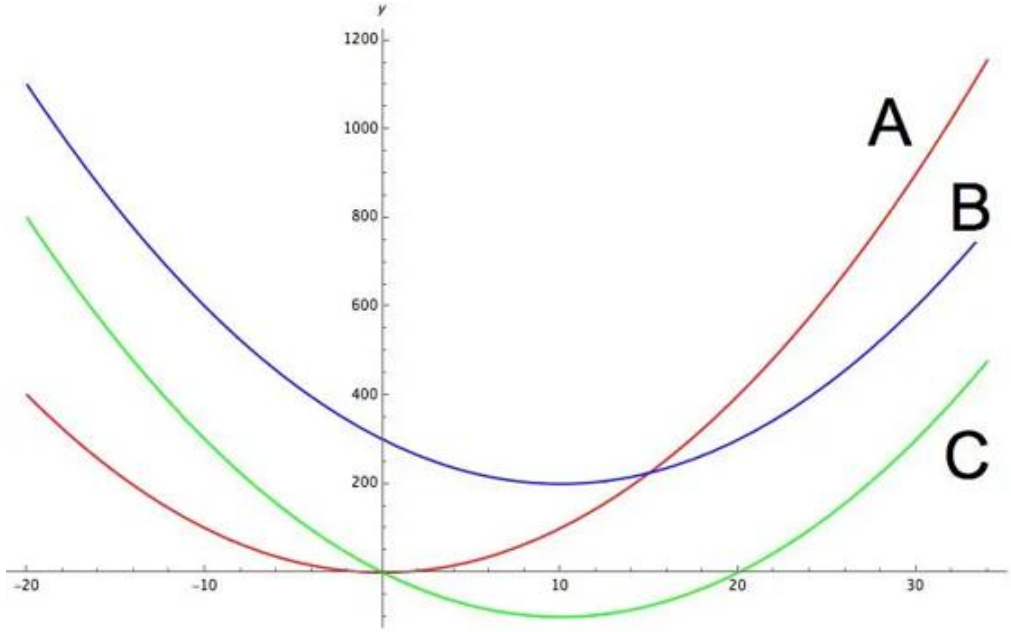
   Answer:

6. Consider two matrices $A$ and $B$. If the dimension of the left nullspace of $(AB)$ is 3 and the dimension of the rowspace of $A$ is 2. What is the minimum and maximum dimension of the range of the linear map represented by $(AB)$.

   Maximum:

   Minimum:

7. The graphs of characteristics polynomials for $2 \times 2$ diagonalizable matrices $A, B, C$ are shown below. (note that the graph for $A$ just touches the x-axis at the origin)




What are the ranks of $A, B, C$ ?

Answer:

8. Let $A_i, i = 1,\ldots,10$ be 3x3 matrices randomly chosen. Briefly explain why there always exist $c_i, i = 1,\ldots,10$ such that not all of them are zero and $\sum_{i=1}^{10} c_i\, A_i = 0$.

Answer:

9. Find all singular values of an SVD of the matrix [1 2 3 4 5].

Answer:

10. What is wrong with this proof? If $\lambda$ is an eigenvalue of $A$ and $\mu$ is an eigenvalue of $B$, then $\lambda\mu$ is an eigenvalue of $AB$ Proof: $Ax = \lambda\boldsymbol{x}$ and $B\boldsymbol{x} = \mu\boldsymbol{x}$ then $AB\boldsymbol{x} = A\mu\boldsymbol{x} = \mu A\boldsymbol{x} = \mu\lambda\boldsymbol{x}$

Answer:

11. Find the pseudo inverse of $\begin{pmatrix} 81 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 25 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Answer:

12. Find the point on the subspace $V = \{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x + z = 0 \}$ that is closest to the point $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Answer:

Part III: (50 points) Show how to solve in detail.

1. (10 points) Let $D_1$ and $D_2$ be two different bases of $\mathbb{R}^n$ and let $B_1$ and $B_2$ be two different bases of $\mathbb{R}^m$. Let $A = \begin{bmatrix} v_1 & \cdots & v_m \\ | & \cdots & | \end{bmatrix}$ be $\mathrm{Rep}_{\mathrm{B}_1,\mathrm{D}_1}(h)$ where $h: \mathbb{R}^m \to \mathbb{R}^n$ is a linear transformation. Describe how to calculate $\mathrm{Rep}_{\mathrm{B}_2,\mathrm{D}_2}(h)$ when we know only $A, B_1, B_2, D_1$ and $D_2$ but not $h$.

2. (10 points) Consider $v_1, v_2$ and $v_3$ which are vectors in $\mathbb{R}^3$. Assume that the result of Gram-Schmidt $(v_1, v_2, v_3)$ is $v_1, w_2, v_3$, i.e., the process of Gram-Schmidt only changes the second vector. There is a transformation $T: \mathbb{R}^3 \to \mathbb{R}^3$ such that both $< w_2 >$ and $< v_2 >$ are two different bases of its null space. We also know that $v_1 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 0 \\ -10 \end{bmatrix}$. Calculate any matrix A such that its column space is the intersection between $span(\{w_2, v_3\})$ and the row space of $T$ .

3. Find all square roots of the matrix

$$\begin{bmatrix} 1 & 3 & -3 \\ 0 & 4 & 5 \\ 0 & 0 & 9 \end{bmatrix}$$

4. (10 points) Let $A$ be a 3x3 matrix such that $A\begin{bmatrix}0\\1\\-1\end{bmatrix}=\begin{bmatrix}4\\0\\2\end{bmatrix}$, the RRE of $A^{\mathsf{T}}$ is $\begin{bmatrix}1 & -3 & -1\\0 & 2 & 1\\0 & 0 & 0\end{bmatrix}$, and the RRE of $A$ is $\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix}$. Given further that an eigenvector of $A$ is $\begin{bmatrix}1\\1\\1\end{bmatrix}$ which is orthogonal to another eigenvector. Find all eigenvalues and their corresponding eigenvectors of $A$.

5. (10 points) Let $T: \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with its nullity equal to zero. If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a linearly independent subset of $\mathbb{R}^n$, show that $\{T(\mathbf{x}_1), T(\mathbf{x}_2), \dots, T(\mathbf{x}_k)\}$ is a linearly independent subset of $\mathbb{R}^m$.